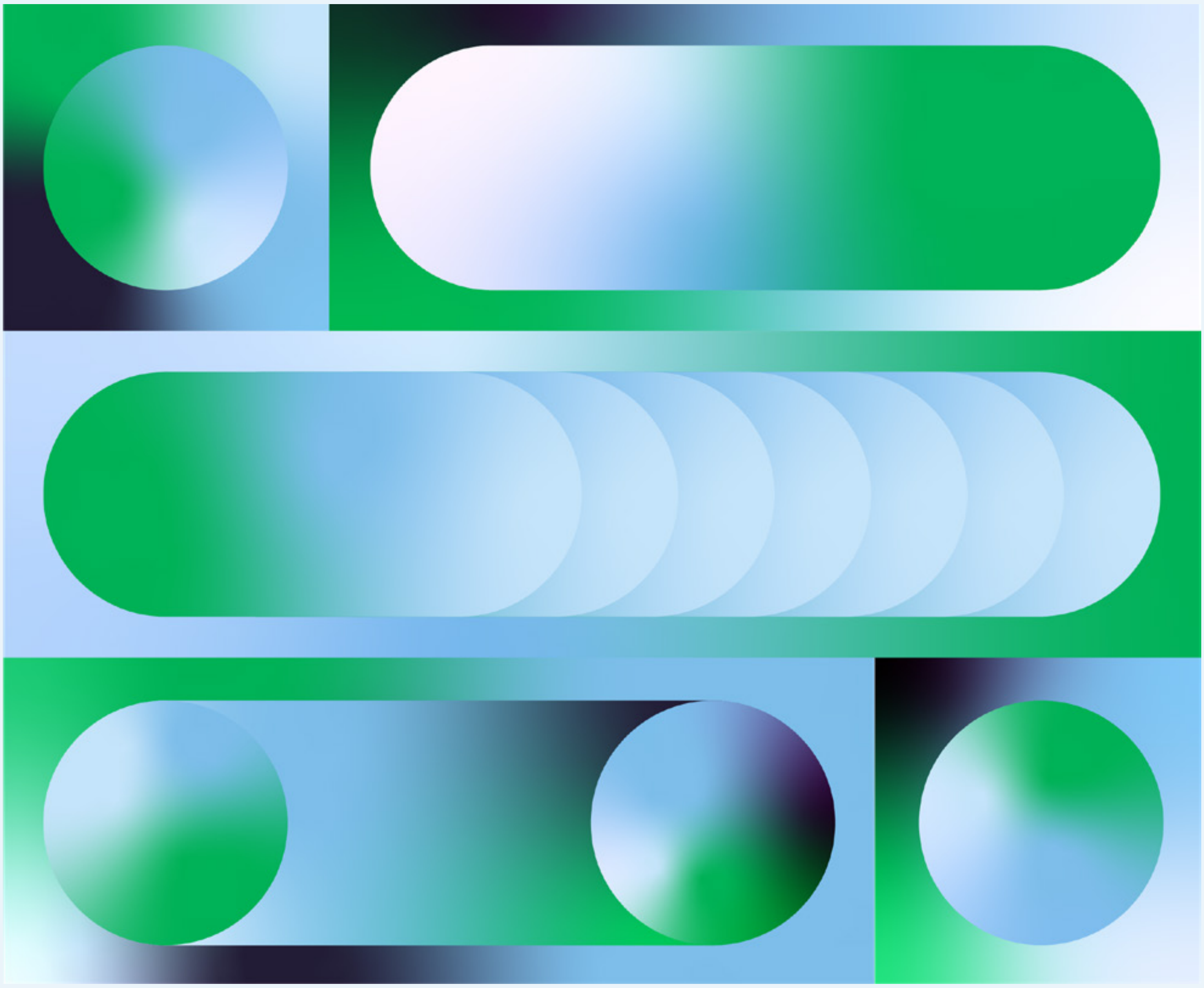
E-MAGAZINE

AI in 2024: Hot Takes From Dataiku, Deloitte, & Snowflake



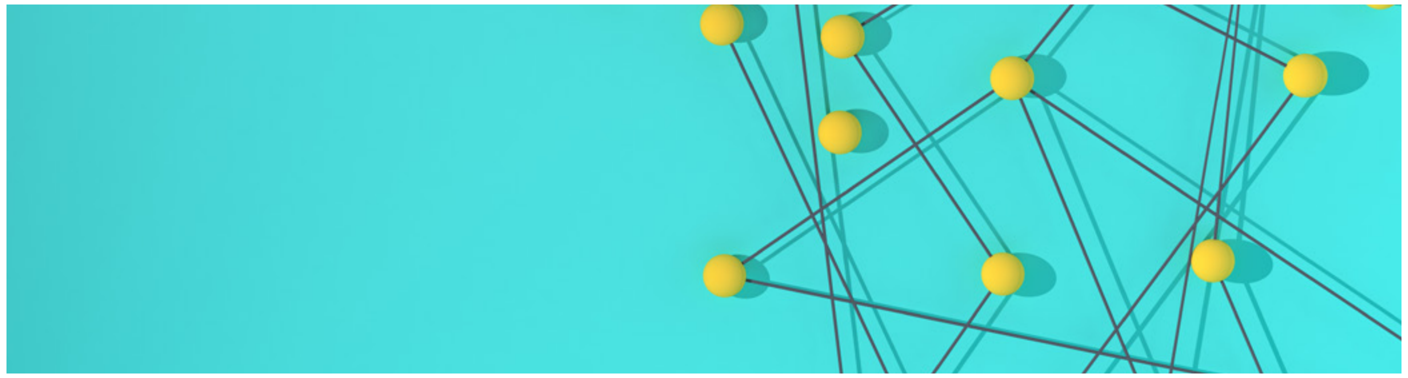
A Coming Era of Compute Cost Volatility

Florian Douetteau
Bio

CO-FOUNDER AND CEO, DATAIKU

Florian Douetteau is the Chief Executive Officer and co-founder of Dataiku, the platform for Everyday AI, enabling data experts and domain experts to work together to build data into their daily operations, from advanced analytics to Generative AI. More than 600 companies worldwide use Dataiku, driving diverse use cases from predictive maintenance and supply chain optimization, to quality control in precision engineering, to marketing optimization, Generative AI use cases, and everything in between.

Florian started Dataiku in 2013 out of his passion for data, machine learning, and people. He envisioned a future for businesses with AI becoming mainstream through the collaborative effort of everyone in the company, not just data scientists or technical experts.



To underscore how AI will become increasingly essential to businesses in every sector, I thought it was best summarized by Mustafa Suleyman, co-founder and CEO of Inflection AI and co-founder of DeepMind (which has since been acquired by Google), in an article for TIME entitled, "How the AI Revolution Will Reshape the World":[1]

> **We are about to see the greatest redistribution of power in history.**
>
> Over millennia, humanity has been shaped by successive waves of technology. The discovery of fire, the invention of the wheel, the harnessing of electricity — all were transformational moments for civilization. All were waves of technology that started small, with a few precarious experiments, but eventually they broke across the world. These waves followed a similar trajectory: breakthrough technologies were invented, delivered huge value, and so they proliferated, became more effective, cheaper, more widespread and were absorbed into the normal, ever-evolving fabric of human life.
>
> **AI is different from previous waves of technology because of how it unleashes new powers and transforms existing power.** This is the most underappreciated aspect of the technological revolution now underway. While all waves of technology create altered power structures in their wake, none have seen the raw proliferation of power like the one on its way."

If this statement is not evidence enough that the ability to augment and automate business processes and business decisions with AI will increasingly become a core capability of every business, this might: **According to Bank of America Global Research and IDC, global revenue associated with AI software, hardware, service, and sales will likely grow at 19% each year, reaching a staggering $900 billion by 2026, compared with $318 billion in 2020.**[2]

However, to achieve the ideal future state, companies will require various resources, notably:

- Access to the skills (and software) to build AI-powered systems
- Access to the Graphics Processing Units (GPUs) that are needed for the use of AI models, notably Large Language Models (LLMs)

Alongside subject matter experts from our trusted partners at Deloitte and Snowflake, I'm going to share my personal take about AI in 2024: We are entering an era of compute cost volatility. GPU costs will be highly variable and difficult to predict and costs will be driven, fundamentally, by supply and demand.

Read on for what this means for the organizations poised to harness AI (including Generative AI) at scale in the months and years to come.

---

1 *https://time.com/6310115/ai-revolution-reshape-the-world/*
2 *https://business.bofa.com/en-us/content/economic-impact-of-ai.html#footnote-2*

"
We are about to see the greatest redistribution of power in history.

The GPU Advantage in Mastering LLMs

As a technical refresher, GPUs are computer processors that are designed to accelerate the calculations necessary to render images and videos on a display. Originally developed for rendering graphics in video games, GPUs have evolved to become highly parallel processors capable of handling a large number of calculations simultaneously.

This parallel processing capability makes GPUs well-suited for certain types of computations, such as those involved in machine learning (ML), deep learning, and AI. Before GPUs came on the scene in the late 1990s and early 2000s, Central Processing Units (CPUs) performed the calculations necessary to render graphics. While both GPUs and CPUs are essential components of modern computing systems, they are optimized for different types of tasks.

LLMs are trained on large datasets and can generate human-like text based on the inputs they receive in a process known as inference. This process is most efficiently done on GPUs, which are adapted for parallel processing (meaning the architecture of these models allows for the simultaneous processing of multiple tokens or words).

The parallel processing capabilities and computational power of GPUs are instrumental in the training and deployment of LLMs — they contribute to faster training times and more efficient inference.

So, for organizations aiming to achieve Everyday AI — building Generative AI as well as traditional ML and advanced analytics into the processes and operations throughout their business — this has implications: The costs associated with GPUs will exhibit significant variability and prove challenging to anticipate, primarily influenced by the fundamental dynamics of supply and demand.

That demand will increase (64% of senior AI professionals are “Likely” or “Very Likely” to use Generative AI for their business over the next year, **according to a Dataiku and Databricks survey**), though we don’t know by how much or how quickly. Supply is uncertain, but will undoubtedly be influenced by manufacturing capacity within the semiconductor supply chain (which includes the production of GPUs and is concentrated in a few locations). In turn, manufacturing capacity may be influenced by global events such as natural disasters or geopolitical events.

How to Prepare for & Manage the Cost Variability

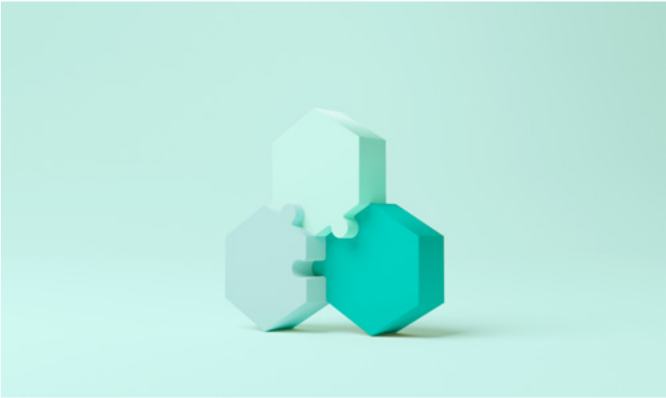
Many companies are experienced in managing variable costs of commodities. For example, companies in energy-intensive industries already do this for energy costs, balancing different energy sources to achieve the right balance between availability, price and, increasingly, carbon intensity. Companies that ship products do this with shipping costs, which are tightly coupled with energy costs.

Where compute cost volatility differs is companies that are not used to this kind of work will need to do it now, as GPU cycles become critical to their work. For example, financial services companies or pharmaceutical companies don't usually engage in energy and shipping trading, but they will need to build such practices for GPU compute. This evolution of compute cost dynamics leads to several consequences:

> **"Companies will need the ability to change between models and service providers to adapt cost posture in the face of variable costs.**

#1

Companies will need the ability to analyze and assess the tradeoffs between cost and quality of output to strike the most effective balance. This will require a nuanced understanding of the variables at play, ensuring that computational resources are deployed efficiently.

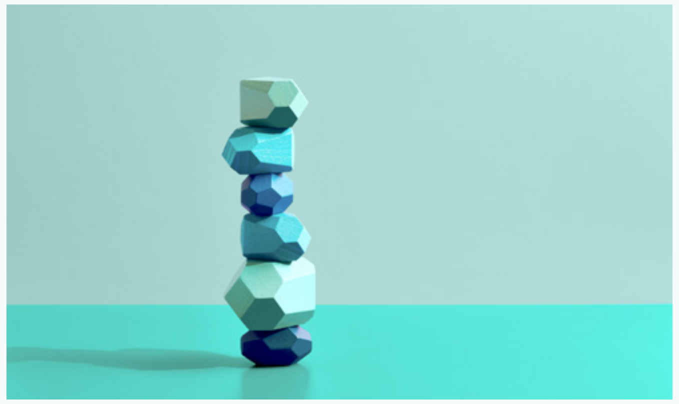


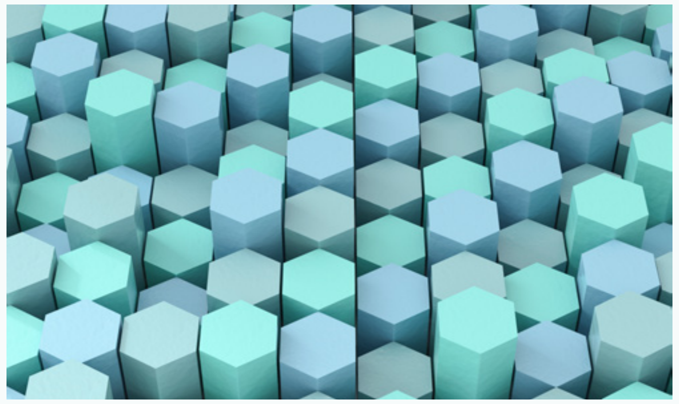
#2

Companies will need the ability to change between models and service providers to adapt cost posture in the face of variable costs. They should also be sure to use automated scaling solutions to dynamically adjust computing resources based on workload demands, which ensures optimal performance during peak times and minimizes costs during periods of lower demand.

#3

As a way to lock in their costs, this shift may drive companies to manage their own GPU clusters, rather than renting them from cloud providers. While this approach introduces the additional overhead of maintaining these clusters, it provides organizations with greater control over their computing infrastructure, potentially offering long-term cost benefits.





#4

Technologies that optimize the models for specific use cases will become more popular. Organizations are increasingly seeking solutions that enhance the efficiency of their GPU utilization, aligning computational resources with the unique requirements of their applications.

The GPU Paradigm Shift: How Dataiku Can Help

Dataiku — the platform for Everyday AI — complements organizational strategies for managing compute costs in the context of the Generative AI boom. With robust orchestration across data science and ML workflows (from data preparation to MLOps), organizations can optimize resource utilization and reduce unnecessary compute costs.

The ability to scale horizontally and vertically enables Dataiku to accommodate varying workloads and demands. This scalability enables organizations to adapt to changing compute requirements without major disruptions, thus helping to manage costs effectively.

Next, efficient data management and feature engineering are essential for building high-performing ML models. Dataiku provides tools for managing and processing data, contributing to the creation of optimized datasets that can lead to more efficient model training, potentially reducing overall compute costs.

Further, the collaborative environment of Dataiku allows data scientists and ML engineers to experiment with different models and algorithms, which can lead to more efficient model development. This can enable organizations to choose models that provide a good balance between performance and computational requirements, thus influencing overall compute costs. Next, Dataiku's visual interface enables users to monitor and optimize the performance of ML models, which includes identifying resource-intensive tasks and optimizing them for efficiency.

With the ability to integrate with leading cloud platforms, users can take advantage of cloud services with elastic computing capabilities, paying only for the resources they consume. In times of compute cost volatility, the ability to scale up or down in the cloud can be a cost-effective strategy.

Plus, organizations can capitalize on cloud-specific features such as auto-scaling to optimize compute costs and ensure seamless deployment and scalability of machine learning models in cloud environments. To build trust in AI projects and programs, Dataiku enables organizations via teamwork and transparency in a single, centralized workspace, explainability and safety guardrails, and robust AI Governance and oversight.

Conclusion

Particularly in the age of Generative AI, teams will continue to generate and accumulate more data, thus the need for computational power to process and analyze this data will increase. So, while organizations should be mindful of potential compute cost volatility, it shouldn't deter them from continuing to scale ML and AI initiatives.

By adopting a strategic and measured approach (such as with Dataiku), organizations can harness the benefits of these technologies while managing and optimizing computational resources effectively.

Deloitte.

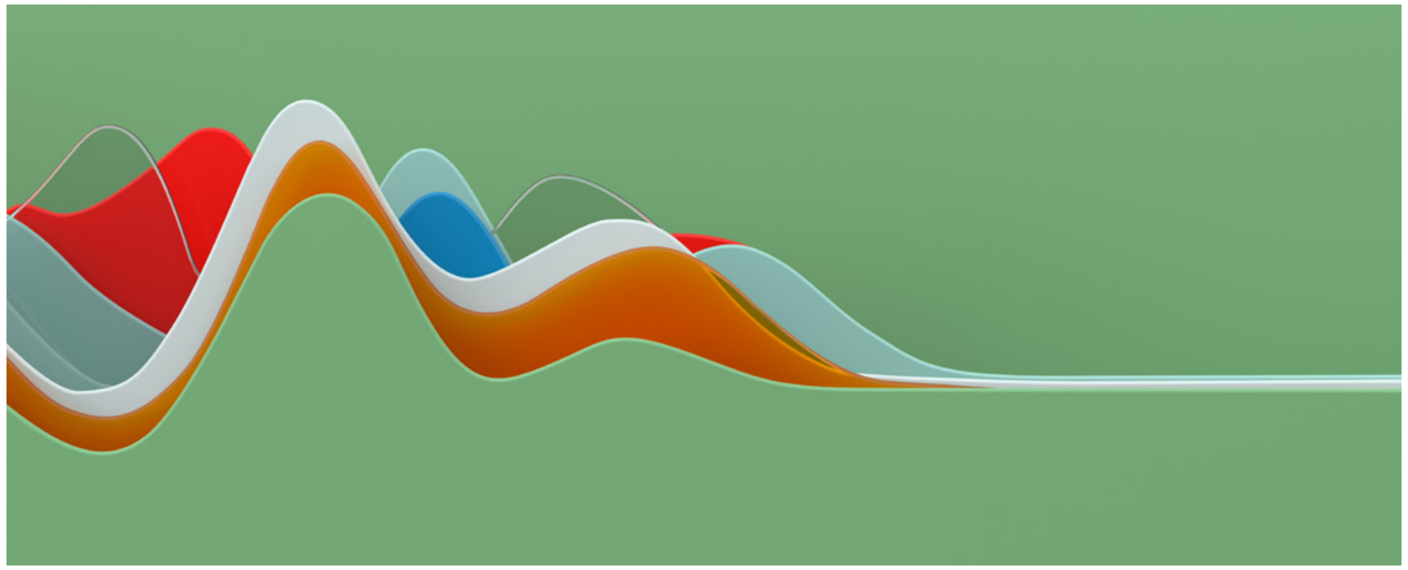
An Opportunity to Build Trust

Oz Karan
Bio

PARTNER, DELOITTE RISK & FINANCIAL ADVISORY, DELOITTE

Oz is a partner within Deloitte & Touche, LLP's Risk and Financial Advisory (R&FA) practice and serves as R&FA's Trustworthy AI leader. He has more than 20 years of risk management, regulatory compliance, and financial advisory consulting services experience with in-depth knowledge of the banking, payments, and fintech industries.

Oz supports clients as an advisor to the Risk, Regulatory, and Finance functional leaders on strategic risk management, regulatory compliance, and operational and technology modernization. He routinely advises client leaders around the possibilities of modern risk & compliance organizations, enabling the organization's most strategic priorities with more effective and efficient processes, governance structure, and technologies.



The age of AI is upon us with the dawn of a broad spectrum of Generative AI capabilities available to consumers. But much like the technologies themselves, trust in these paradigm-shifting technologies will not be built in a day.

As with prior cutting-edge tech, AI is greeted with a healthy skepticism from both organizations and consumers seeking assurances on the privacy and security of their data. This skepticism is further compounded by the black box these solutions can present to AI operators, owners, and developers. As organizations grapple with their use of AI, the line between trust in the machine and trust in the organization blurs.

As AI solutions quickly integrate into many facets of everyday life, consumers and employees won't distinguish between trust in an organization and trust in its use of AI, or its AI output. This will make AI a strategic operational opportunity and a core tenet of brand and reputation management.

The question of "Whom can I trust?" is one we ponder when assessing another's character, intent, or motivations. Personal trust is carefully cultivated over time; it requires consistency, familiarity, and prioritization of meeting that individual's needs.

So, when faced with a technology solution that obfuscates the 'how' and 'why' of its outputs while introducing questions like "Is my data the product?", AI users must prioritize establishing and continually proving trustworthiness.

At Deloitte, we work to understand the risks, rewards, utility, and trustworthiness of AI so that we may help clients across industries, government, and commerce leverage the technology. Deloitte's Trustworthy AI™ Framework provides a cross-functional approach to help organizations identify key decisions and methods needed to maximize Generative AI's business potential.

We know from Deloitte's TrustID Generative AI study of over 500 respondents that consumer trust decreases by 144% when consumers know a brand uses AI[1], and that their perception of reliability drops by 157%[1]. Similarly, employees' trust in their employers decreases by 139% when employers offer 'AI technologies' to their workforce[1].

Human trust in AI is an uphill climb. The organizations that focus on building AI solutions with trust by design may claim an advantage in the marketplace.

1 *Deloitte's TrustID Generative AI Analysis. August 2023.*

The age of AI is upon us with the dawn of a broad spectrum of Generative AI capabilities available to consumers. But much like the technologies themselves, trust in these paradigm-shifting technologies will not be built in a day.

> **The velocity of AI solutions' availability necessitates balancing innovation with forward-thinking control mechanisms and technological guardrails. Trustworthy AI deployment will likely require a rebalancing of accountabilities and responsibilities across the organization, necessitating cross-functional relationships between operations and technology.**

Understanding Trust Erosion Possibilities

AI is not infallible. Countless examples in the public sphere have proved as much from rogue text bots to intellectual property infringement to data exfiltration. Understanding and accepting that AI may fall short of human expectations can help limit the consequences of those failures and formulate responsible planning and response. If for humans, seeing is truly believing, it may be an onerous journey for AI to gain human trust.

Consider this example: Waymo's joint study with Swiss Re insurance found the driverless rideshare company's cars experience 76% fewer accidents involving property damage compared to human-driven cars[2]. But comparative statistics alone do not shape public perception, and the perceived risk of an AI solution often pales in comparison to its perceived reward for widespread adoption.

The velocity of AI solutions' availability necessitates balancing innovation with forward-thinking control mechanisms and technological guardrails. Trustworthy AI deployment will likely require a rebalancing of accountabilities and responsibilities across the organization, necessitating cross-functional relationships between operations and technology.

Usage of AI and accountability should be driven from the top down, with boards and management answering for AI incidents similarly to cyber events or regulatory noncompliance. Management that cannot or does not effectively speak to the organization's use of AI could risk fostering a culture of nonaccountability.

Both consumers and employees expect organizational transparency when it comes to the use of AI, and the use of their data in AI. As regulators trend toward requiring informed consent from data subjects, organizations will need to both comprehensively understand and communicate their AI uses, not simply for regulatory compliance purposes, but to keep the lines of communication open with crucial stakeholders inside the organization and out.

2 *Comparative Safety Performance of Autonomous- and Human Drivers: A Real-World Case Study of the Waymo One Service*

Building AI Solutions With Trust by Design

Across the globe, regulators have identified common themes and core characteristics for organizations to consider as they develop and deploy AI. In the U.S., the White House established an AI Bill of Rights[3] highlighting its AI priorities, followed by the National Institute of Standards and Technology (NIST) providing guidance to organizations in understanding, assessing, and managing risk with its own Artificial Intelligence Risk Management Framework[4].

This early regulatory guidance provides an outline for proactive organizations to begin constructing guardrails for responsible AI usage. Most recently at the federal level, the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence[5] seeks to establish safety testing for specific dual-use foundational models to identify and mitigate national security AI risks.

The Executive Order also encourages other regulatory bodies to exercise their existing powers to investigate bias, fairness, privacy, and other potential harms rendered by the use of AI, much like some state-level regulations have for specific industries.

With the full weight of regulation still unknown, organizations should continue paying diligent attention to new rules as well as the application of existing regulations. Though hard to imagine, the complexity of today's AI technologies will likely pale in comparison to the solutions to come.

Organizations that wrangle with the challenges of establishing trustworthy AI design practices today can set themselves up for success into the future.

So, what are trustworthy AI design practices? *"Trustworthy AI"* refers to the leading practices and ethical standards by which humans design, develop, and deploy the technology that makes it possible for AI machines to produce safe, secure, and unbiased outputs. This is different from *"trust in AI,"* which is a deeper, intrinsic trust between human and machine.

To establish *"trust in AI,"* organizations must embed knowledge, responsibility, and accountability to uphold organizational values and the trust afforded from consumers, employees, communities, and other constituencies. Organizations that articulate and highlight how they secure trust using AI technology in the right ways may have an easier path to gaining people's trust in AI.

Those who continue to build and refine today's AI platforms must keep trust top of mind as they research and develop the next generations of Generative AI. Think of it as "trust by design," where AI designers weave the aspiration into the very fabric of the technology, with all regulatory expectations for safety, security, and accountability firmly in mind.

3 *National Institute of Standards and Technology AI Risk Management Framework*

4 *The AI Bill of Rights follows the Executive Order 13960: Promoting the Use of Trustworthy Artificial Intelligence in the Federal Government (December 2020).*

But How to Inspire Trust?

To further balance AI innovation with adequate control mechanisms, consider the value of establishing an AI Governance framework that's aligned to your organization's corporate values. Adhering to regulations and aligning with the organization's values will help demonstrate responsible stewardship of AI both to employees and the public.

By establishing a greater focus on AI controls, including third-party risk management, organizations can build the credibility they crave to form a bedrock of trust.

A foundation of trust is slowly built upon transparency and easily damaged by omission. To effectively establish transparency regarding AI solutions, organizations can proactively share the results of safety testing, such as red-team testing called for in the recent Executive Order[5]. Explaining to core constituencies the types of guardrails employed to guard against potential harms of AI can help protect consumers[6] and establish public trust.

5 *Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (October 2023)*

6 *Stanford University Human-Centered Artificial Intelligence 2023 Foundation Model Transparency Index*

The Rewards of Trust

AI and Generative AI are not domains which will be won or lost. The question is which organizations will win *with* AI? As we enter the *Age of With*[7], trust can be the differentiator between success and failure in this technological revolution.

By building solutions that incorporate trust into all stages of AI development and use, organizations stand to benefit both themselves and society. Organizations that can continually demonstrate prioritization of their key stakeholders through a transparent approach to adoption and effective use of guardrails will continue to gain the trust of the public.

Organizations today know AI is an imperative — those that deem trust of equal criticality will help realize the benefits of AI for all.

7 *Deloitte's The Age of With™ Exploring the future of artificial intelligence*

"AI and Generative AI are not domains which will be won or lost. By building solutions that incorporate trust into all stages of AI development and use, organizations stand to benefit both themselves and society.

snowflake

The Future of Advanced AI Is Simple

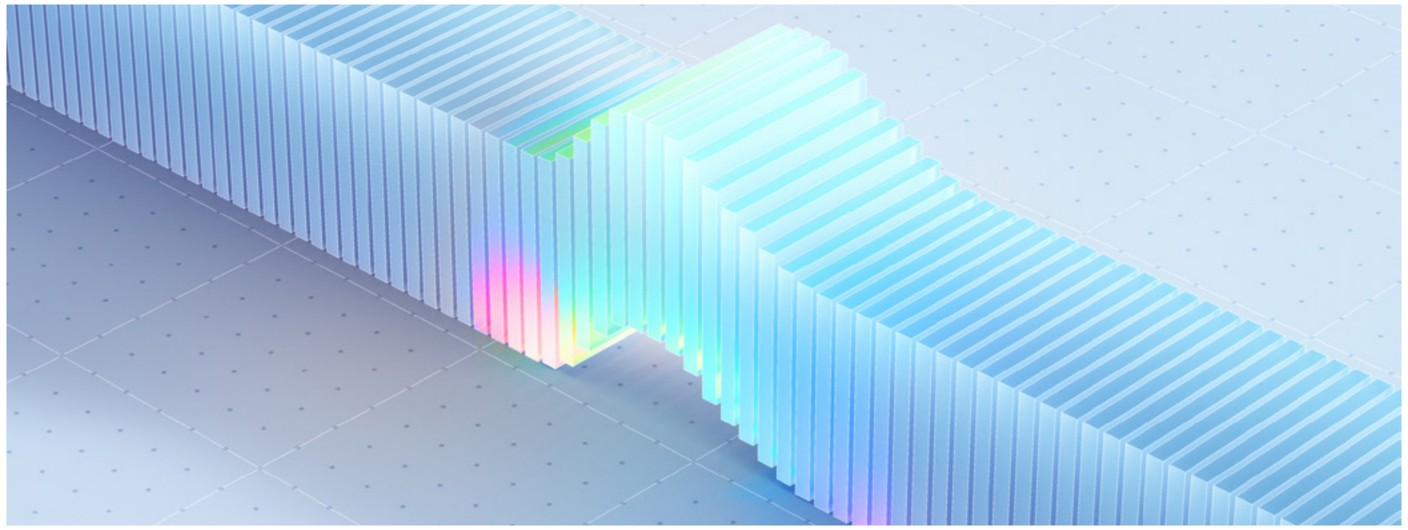
Ahmad Khan

Bio

HEAD OF AI/ML STRATEGY, SNOWFLAKE

Ahmad is the Head of AI/ML Strategy at Snowflake where he helps customers optimize their ML workloads on Snowflake. He also works closely with the Snowflake product team to help define the AI feature set within Snowflake based on the voice of the customer.

Prior to Snowflake, Ahmad spent over four years at AWS where he focused on the AWS stack of ML services and was involved in early proof of concepts for AWS SageMaker. Ahmad holds a Master's in Electrical & Computer Engineering from University of Southern California.



In the age of Generative AI, the easy access to cutting-edge models is quickly setting the stage for enterprises to realize that the real differentiator is their proprietary data and the models that are customized or fine-tuned with that data. At the same time, Generative AI is quickly making advanced AI accessible beyond highly technical data scientists by making the interface between humans and digitized information to be natural language rather than code.

The combination of these two trends means it is critical for enterprise leaders to establish a solid foundation for data and custom models as well as define a strategy that focuses on securely delivering AI in the form of natural-language-oriented application interfaces.

Build and Deploy LLM Apps in Minutes

Similarly to other AI processing such as machine learning models used for predictive analytics, Generative AI demands large volumes of data processing with specialized compute environments.

The common approach to advanced processing has been to enable AI teams to gain access to the enterprise data and make copies of it to use on their own platforms.

This approach to move data from its governed source has caused pain points by creating new silos and proved to be a less than ideal process for security teams. These teams need to analyze vulnerabilities as data gets shuffled across a wide range of compute environments that process data and serve model results.

A modern approach is to enable the data scientists and other developers to do their advanced processing where the data is already curated and governed. By reducing data movement, developers can both iterate faster as part of their development cycle and reduce security and operational complexities to take projects to production.

Built with data-intensive processing in mind, Snowflake offers scalable infrastructure and Large Language Model (LLM) application stack primitives that enable developers to build apps in just minutes without moving data or creating copies. This includes Snowflake Cortex, which brings access to leading LLMs such as Llama 2, as well as Snowpark Container Services, which supports the execution of models packaged as containers.

> "The combination of these two trends means it is critical for enterprise leaders to establish a solid foundation for data and custom models as well as define a strategy that focuses on securely delivering AI in the form of natural-language-oriented application interfaces.

Use AI in Everyday Analytics Within Seconds

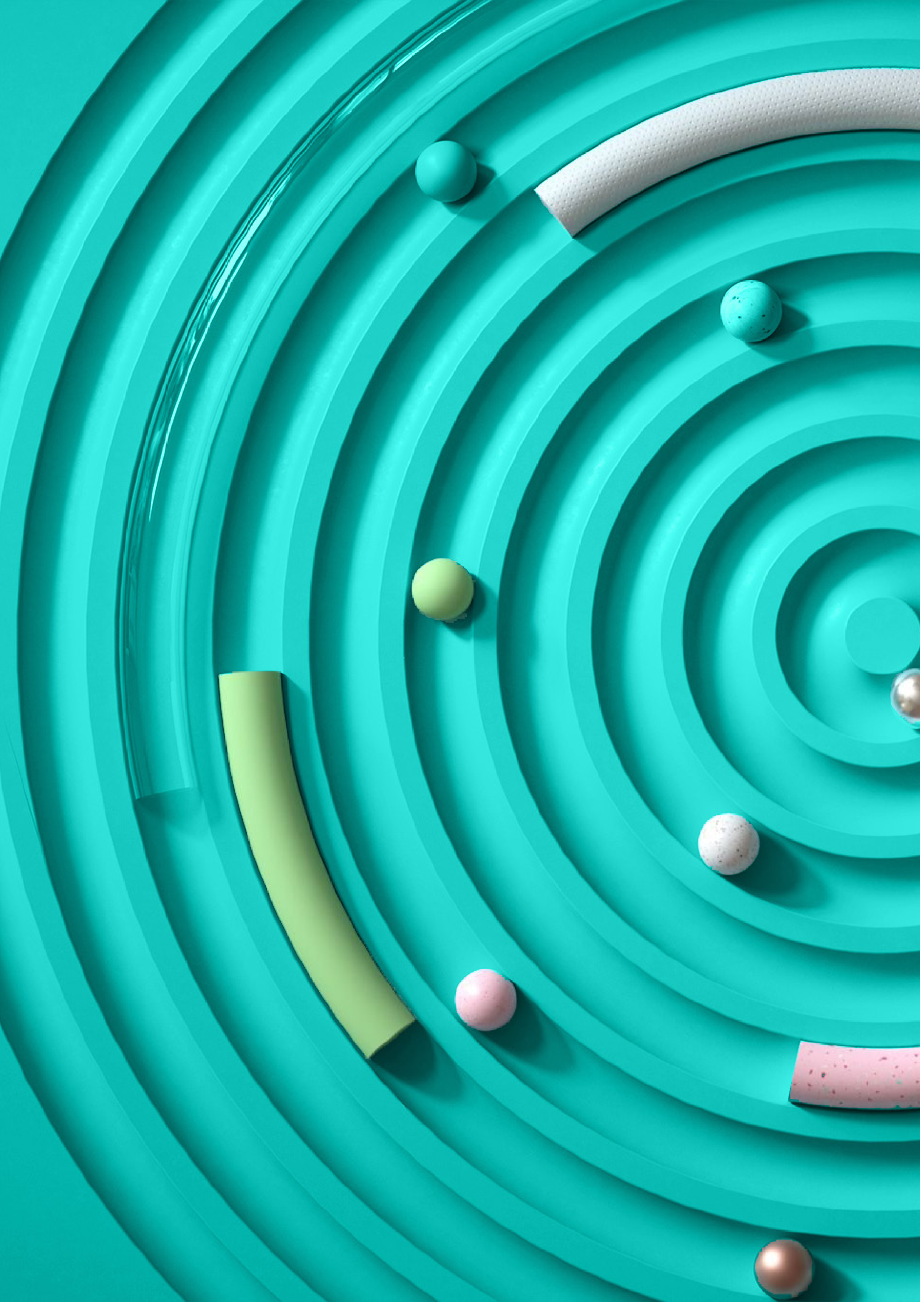
As part of a comprehensive Generative AI strategy, data executives must also identify paths to expand adoption beyond the AI experts and drive innovation among the analysts and business teams.

This could be done by enabling multiple teams without engineering backgrounds to use LLMs. This could include analysts using LLMs as part of familiar SQL functions as well as enabling business teams to use applications with graphical user interfaces that keep data and processing in Snowflake. Snowflake offers first-party applications with pre-built UIs such as Document AI to help non-coders search and get answers from PDF documents.

Getting the most value from Generative AI will require organizations to define a holistic strategy that first establishes a robust data and model governance and then enables developers to accelerate LLM app development and analysts to leverage AI as part of everyday analytics. 2023 was a pivotal year for enterprise AI and we look forward to partnering with more organizations as part of their AI strategy in the coming year.

“

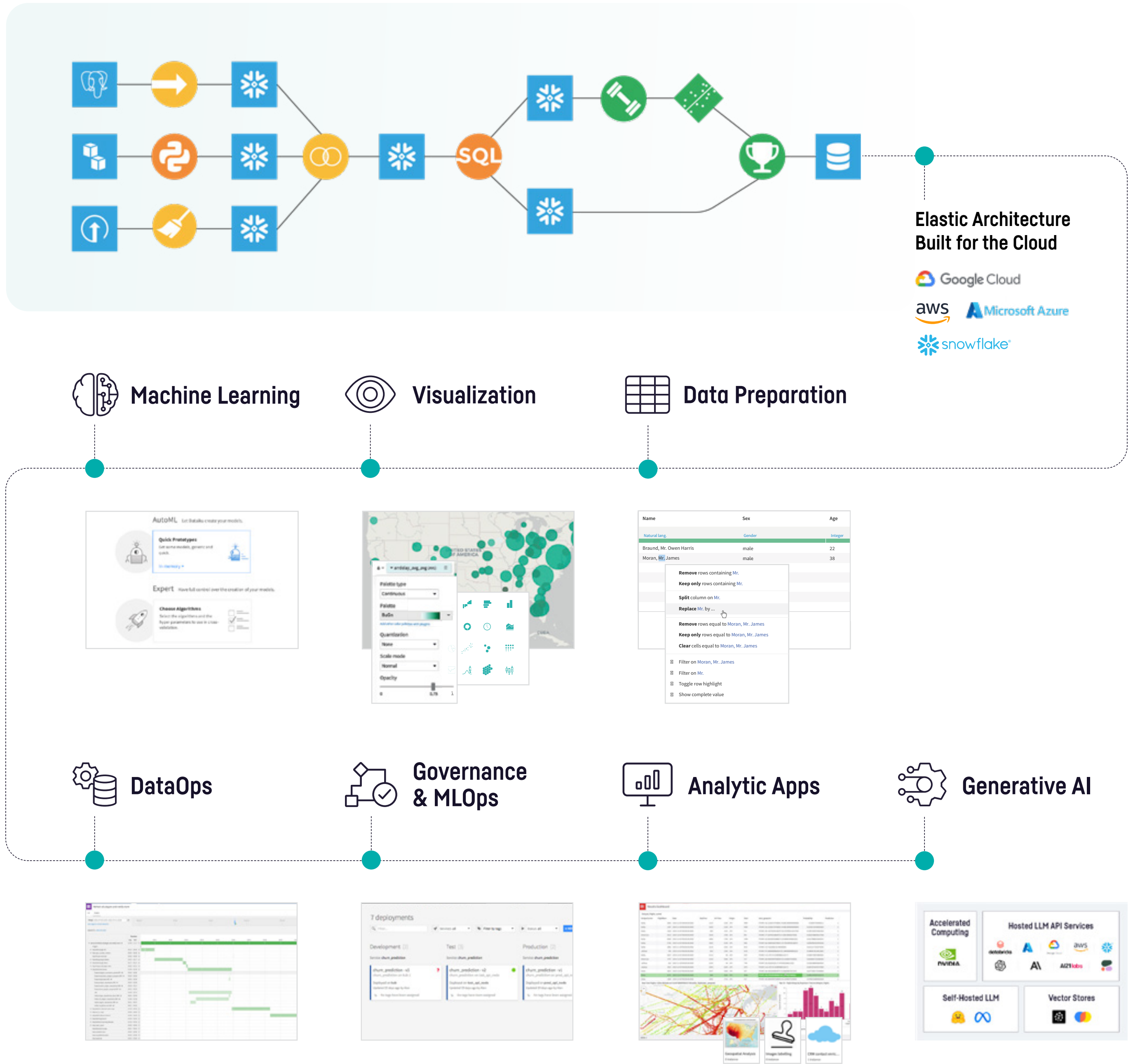
Getting the most value from Generative AI will require organizations to define a holistic strategy that first establishes a robust data and model governance and then enables developers to accelerate LLM app development and analysts to leverage AI as part of everyday analytics.



dataiku

Everyday AI, Extraordinary People

Dataiku is the platform for Everyday AI, enabling data experts and domain experts to work together to build data into their daily operations, from advanced analytics to Generative AI. Together, they design, develop and deploy new AI capabilities, at all scales and in all industries.



dataiku